Multiobjective Memetic Spatiotemporal Subpixel Mapping for Remote Sensing Imagery

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***Abstract*— Subpixel mapping (SPM) technology is an effective way to account for the distribution of the component objects within the mixed pixels and can alleviate the mixed pixel problem to some degree. However, the traditional SPM methods rely on only a single coarse-resolution image, and the limited information source can lead to great uncertainty. The rapid development of Earth observation systems has resulted in many fine spatial resolution remote sensing images now being accessible. Spatiotemporal SPM approaches utilize the finer spatial distribution of a historical thematic map to provide temporal prior information for the mapping process. However, spatiotemporal SPM is essentially a constrained optimization problem that aims to predict the optimal class distribution map subject to the abundance, spatial, and temporal constraints. It is a challenging task to properly model and optimize the multiple constraints. In this article, a multiobjective memetic spatiotemporal SPM (MOMSPM) framework is proposed. This model transforms the data fidelity term, spatial prior term, and temporal prior term into a multiobjective optimization problem (MOP) to discard the sensitive regularization parameters. The multiobjective model realizes the fusion of abundance, spatial, and temporal information. To optimize the three objective functions simultaneously, MOMSPM provides a multiobjective memetic algorithm framework in which the global multiobjective search method [multiobjective evolutionary algorithm based on decomposition (MOEA/D)] combines two commonly employed single-objective local search operators (geospatial distribution preference (GSDP) local search and maximum a posteriori (MAP)-based local search). The GSDP and MAP operators are employed to refine the solution and achieve an improved outcome. The hybridized method provides powerful search ability and achieves a good balance between the three objective functions. Experiments on synthetic and real datasets prove that the proposed method is superior to the state-of-the-art spatiotemporal SPM methods.**

***Index Terms*— Memetic algorithm, multiobjective optimization, remote sensing imagery, subpixel mapping (SPM).**

I. Introduction

REMOTE sensing imagery plays an increasingly important role in our daily lives, due to its wide range of applications. For example, satellite imagery can be used for target tracking [1], [2], environmental monitoring [3], urban planning [4], and so on. However, due to the limitation of the spatial resolution of satellite sensors and the complex diversity of ground objects, the problem of mixed pixels is ubiquitous in remote sensing imagery.

The mixed pixel problem greatly hinders the development of remote sensing Earth observation technology toward a quantitative direction. Spectral decomposition and subpixel mapping (SPM) techniques are often used to resolve the mixed pixel problem. The spectral decomposition technique can acquire the components of mixed pixels (endmembers) and the corresponding fractions (abundances) in a mixed pixel [5]. However, the spectral unmixing technique is unable to localize the spatial distribution of the endmembers within mixed pixels, which can be resolved by the SPM technique.

SPM was first proposed by Atkinson et al. [6] to predict the spatial location of different classes at the subpixel scale in mixed pixels. The SPM technique can help to obtain high-resolution (HR) classification maps with finer spatial distribution details. The traditional SPM methods cut the mixed pixels into smaller units (subpixels) and assign a specific class to these subpixels according to the abundance image and spatial dependency assumption, i.e., subpixels that are spatially closer together are more likely to belong to the same class than those that are farther apart. For example, the spatial attraction-based SPM (SASM) model proposed by Mertens et al. [7] obtains the SPM results by comparing the attraction values of adjacent subpixels and pixels. The pixel-swapping algorithm (PSA) [5] swaps the categories of subpixels within the same pixel to maximize the spatial attractiveness. The Hopfield neural network (HNN) [8] was designed to obtain the SPM result that maximizes the spatial correlation according to the principle of minimizing the energy function. The trained artificial neural network methods learn the spatial pattern through samples to obtain the optimal mapping solution. Mertens et al. [9] first proposed the use of common artificial shapes, such as circles and rectangles, to train a backpropagation (BP) neural network for SPM. Subsequently, Zhang et al. [10] established the BP reconstruction method by combining a BP neural network with the observation model. Furthermore, geostatistical models [11], [12] have also been proposed to describe the spatial distribution patterns of various ground objects and incorporate the spatial distribution information. The spatial correlation can also be regarded as a prior term. For example, the maximum

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a posteriori (MAP) model [13] uses a spatial prior model as prior knowledge to solve the SPM uncertainty. In addition, to mitigate the effect of the ubiquitous point spread function (PSF) in SPM, Wang et al. [14] proposed a general solution that accounts for the PSF impact on the prespectral unmixing process and provides enhanced abundance maps for SPM. Although the above methods have been successfully applied to SPM, they utilize insufficient information from the abundance map. The HR result maps of SPM are obtained only by maximizing the spatial correlation criterion, and the lack of subpixel-level auxiliary spatial information leads to a limited mapping accuracy. To further improve the accuracy of the location results, it is feasible to add auxiliary data to provide more information about the spatial distribution of the ground objects. With the rapid development of Earth observation systems, a large number of multitemporal remote sensing images are now easily accessible.

There is also a strong correlation between temporally close images in the distribution of ground objects. The powerful temporal correlation can serve as temporal a priori knowledge or a temporal constraint for solving the ill-posed SPM problem. Many scholars have conducted studies on spatiotemporal SPM. With the aim to predict the subpixel spatial pattern with bitemporal images, Ling et al. [15] presented a subpixel land-cover change mapping (SLCCM) technique, where the resulting subpixel HR map is determined by comparing the former HR land-cover image with the abundance maps derived from the latter low-resolution (LR) image. He et al. [16] proposed a spatiotemporal SPM algorithm based on the swarm intelligence method (SSM_CS), integrating short-interval temporal information and fine spatial distribution information within a bitemporal image pair. Xu and Huang [17] introduced a spatiotemporal PSA (STPSA) method on the basis of the traditional PSA model. Li et al. [18] proposed a spatiotemporal HNN (STHNN) method. By the use of the former fine-resolution land-cover map, the STHNN method distinguishes the changed and unchanged areas and utilizes the temporal information of unchanged pixels in the HNN. Zhang et al. [19] proposed a cloud-independent spatiotemporal SPM (C-STSPM) technique to reconstruct fine spatial resolution land-cover maps by utilizing fine-resolution but cloudy images acquired at similar times. A distinctive feature of C-STSPM is that it utilizes the usually discarded cloudy images and introduces a cloud-independent spatiotemporal dependence approach to account for the presence of cloudy pixels. Deng and Zhu [20] introduced a technique for continuous subpixel monitoring (CSM) of urban impervious surface changes. CSM utilizes a time-series analysis to estimate the impervious surface area for each pixel and combines this with SPM methodology, which is an approach that has been found to deliver satisfactory results for impervious surface area estimation. He et al. [21] estimated the subpixel spatial distribution based on MAP theory and a spatial regularization model. To restore specific spatial patterns, a historical HR map derived temporal regularization term was proposed and incorporated into a joint spectral–spatial–temporal MAP-based (SST_MAP) model. Zhang et al. [22] adopted a regularization-based framework similar to SST_MAP to reconstruct 25-m resolution forest maps by fusing time-series Moderate Resolution Imaging Spectroradiometer (MODIS) normalized difference vegetation index (NDVI) images and PALSAR/PALSAR-2 data. However, the weight factor of the SST_MAP model is empirically set and quite sensitive, which affects the accuracy of the result map. Furthermore, the weighted combination of abundance constraints, the data fidelity term, the spatial prior term, and the temporal prior term in SST_MAP cannot achieve dynamic fusion and balance.

Multiobjective optimization problem (MOP) theory [23], [24], [25], [26], [27] can successfully solve this problem and can simultaneously optimize several objective functions. Recently, the multiobjective subpixel land-cover mapping (MOSM) method [28] successfully applied an MOP in SPM, in which the abundance constraint and the spatial constraint are optimized by a multiobjective evolutionary algorithm (MOEA) [29], [30], [31], [32]. The inspiration for MOEAs comes from the biological evolution process. By simulating the cooperative mechanism of a biological population, MOEAs have a global optimization ability. MOEAs can optimize several objective items synchronously and can adaptively fuse the prior information contained in the target item. The MOSM method consists of only two terms (i.e., the data fidelity term and spatial prior term) and its search space is relatively small so that the application of MOP theory works well. Differing from the MOSM method, the spatiotemporal SPM methods consider the temporal information from time-series imagery. As a result, the search dimension is larger and the search process is more complex, which makes the degree of uncertainty higher. Therefore, the search efficiency needs to be further enhanced.

Differing from the MOSM model with only two terms (the data fidelity term and spatial prior term) and its simple search space, the spatiotemporal methods introduce a temporal prior term, which makes the degree of uncertainty higher and the search more complex so that it is necessary to improve the optimization ability of the MOP.

A hybrid of evolutionary algorithms with local search, namely, memetic algorithms [33], [34], [35], [36], [37], have proven to be a very promising way to enhance the search ability [38]. These algorithms are also called genetic local search algorithms and can be obtained using multiple local search engines within the framework of specific evolutionary algorithms. A memetic algorithm can improve the search performance of the evolutionary algorithm, accelerate the convergence, and find a set of representative Pareto-optimal solutions [39], [40]. Therefore, in this article, we propose a multiobjective memetic spatiotemporal SPM (MOMSPM) framework, in which the SPM problem is formulated as an MOP. The memetic algorithm provides a powerful search engine for the MOP to obtain the optimal HR result map.

The main contributions of this article can be summarized as follows.

1) A multiobjective model for spatiotemporal SPM is proposed. In the proposed MOMSPM framework, three objective functions (i.e., data fidelity term, spatial prior term, and temporal prior term) are optimized simultaneously. The data fidelity term describes the

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correspondence between the LR abundance maps and the fine subpixel spatial distribution maps. The spatial prior term models the spatial dependence of the subpixels. The temporal term represents the relevance of the historical images to the current images. It is worth noting that the multiobjective model is able to avoid the sensitive regularization parameter problem.

2) A multiobjective memetic algorithm for model optimization is proposed. For the optimization of the multiobjective SPM model, a multiobjective memetic algorithm hybridized with a geospatial distribution preference (GSDP) operator and an MAP-based operator is designed. A set of candidate HR distribution maps form the initial solutions, and an MOEA based on decomposition (MOEA/D) serves as the global search method. While the global search iteratively searches for the optimal mapping solution with uncertainty, the GSDP local operator and MAP-based local operator are proposed for fine-tuning the mapping solution. The GSDP local search can effectively increase the spatial correlation between subpixels, and the MAP-based local search can accelerate the convergence and easily find the local maximum. During the optimization process, the data fidelity term, spatial prior term, and temporal prior term can be dynamically integrated, thus producing a mapping solution with high quality.

The rest of this article is organized as follows. The basic principles related to spatiotemporal SPM are presented in Section II. The framework and details of MOMSPM are illustrated in Section III. We describe the experiments and analysis conducted in this study in Section IV to evaluate the performance of MOMSPM. Our conclusions are finally drawn in Section V.

II. Related Work

A. Problem Description

For single-image SPM, the aim is to generate an HR classification map based on the LR abundance maps. Each coarse pixel in an LR abundance map corresponds to an $S \times S$ subpixel block in the HR classification map. The observation model for SPM between the HR classification map and LR abundance map of class c can be expressed as follows:

$$y^c = Dx^c + n^c, \quad 0 \le x^c \le 1 \tag{1}$$

where y^c is the LR abundance map of class c. The downsampling matrix D is related to the mapping scale S. x^c is the corresponding HR classification map regarding class c in which the values range from 0 to 1. For every coarse pixel of y^c, Dx^c represents the average value of the corresponding $S \times S$ subpixel block inside x^c. n^c is the potential noise.

However, SPM is an inverse ill-posed problem. Most of the traditional SPM methods are based only on the spatial dependence assumption and take a single image as the research object. The limited information from a single image and the inadequate subpixel spatial auxiliary information lead to ambiguous results, as shown in Fig. 1. When the mapping scale is set to 2, the central pixel in the abundance map is

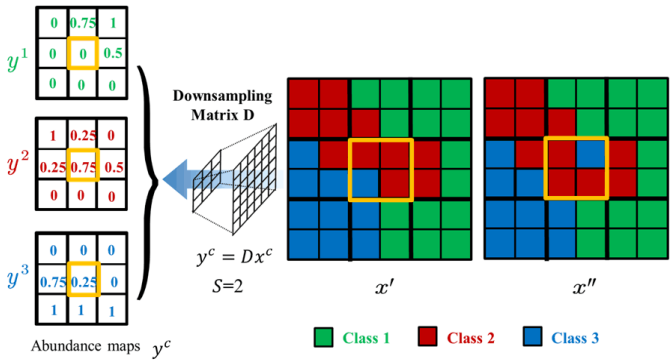


Fig. 1. Problem of SPM.

divided into four subpixels. It is difficult to distinguish which of the two possible spatial distribution solutions x'' and x'' (both satisfy the abundance constraint) is better. The utilization of abundance information alone is not sufficient to locate the optimal subpixel distribution.

Therefore, according to the first law of geography, in that everything is related to everything else and near things are more related than distant things [41], the MAP model [13] introduces spatial prior information into the SPM method. It transforms the ill-posed problem into a regularization problem, for which the data fidelity term and the spatial prior term are based on abundance constraints and spatial correlation constraints, respectively. With the mapping model (1), the MAP model regularizes the ill-posed SPM problem to be well-posed. The minimization cost function of MAP-based SPM is defined as

$$x^c_{\text{MAP}} = \arg\min\left\{ \left\| y^c - Dx^c \right\|^2 + \lambda U\left(x^c\right) \right\}. \tag{2}$$

In this model, $\|y^c - Dx^c\|^2$ is the data fidelity term that controls the quantitative relationship between the HR distribution images and the LR abundance images, and $U(x^c)$ is the spatial prior term, describing the spatial distribution pattern of class c in image x^c. The Laplacian model [42], total variation (TV) model [43], and bilateral TV (BTV) model [44] are common spatial prior models. λ is the weight regularization parameter, which can keep a balance between the data fidelity term and the spatial prior term; however, it is quite sensitive.

B. Multiobjective Optimization

The MOP is very applicable to a problem involving several objective functions that need to be minimized or maximized. These objective functions may mutually conflict. The model of an MOP can be established as shown in the following:

$$\begin{aligned} &\min\ F(x) = (f_1(x), \ldots, f_m(x)) \\ &\text{s.t. } x \in G \subseteq R^n \end{aligned} \tag{3}$$

where x is an n-dimensional decision vector and G is the feasible set in decision space. $f_i(x)(i = 1, 2, \ldots, m)$ is the ith objective function. If we suppose that $u, v \in R^m$, when (4) can be satisfied, v dominates u

$$\forall i \in [1, m] \quad f_i(u) \le f_i(v), \quad \exists f_i(u) \ne f_i(v). \tag{4}$$

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Given a point x^* in the feasible set G, x^* can be named as nondominated or Pareto-optimal when no other point in the set can dominate it. $F(x^*)$ is named a Pareto-optimal vector. Any small improvement in one objective of a Pareto-optimal point can cause a deterioration of the other objective functions [35]. The set containing all the Pareto-optimal points (or all the Pareto-optimal objective vectors) is usually named the Pareto set (PS) or the Pareto front (PF).

The purpose of the MOP is to acquire a PF with high quality. In the past, many scholars have proposed different methods to resolve MOPs. In the proposed approach, a memetic algorithm hybridizing two local search operators with the global search method (MOEA/D [28]) is used.

III. Proposed Method

The MOMSPM framework includes two main parts: 1) the establishment of the multiobjective spatiotemporal SPM model and 2) the optimization of the proposed model. In the first part, the three objective functions derived from the SST_MAP method are introduced. In the second part, multiobjective memetic optimization is introduced, combining MOEA/D with two local operators. The flowchart of the proposed MOMSPM algorithm is shown in Fig. 2. The details of the proposed framework are given as follows.

A. Multiobjective Model for Spatiotemporal SPM

Recently, the rapid development of Earth observation systems has made it possible to acquire time-series remote sensing images. There can be a strong correlation between the time-series images and the fine spatial patterns and details in temporally close images, which can provide valuable temporal information. Therefore, there is a certain similarity between the class distribution of temporally close images, which can be modeled as a temporal prior term, similar to the way that the spatial correlation is constructed as the spatial prior term.

The combination of the data fidelity term, spatial prior term, and the temporal prior term is a good way to produce optimal SPM solutions. When the data fidelity term and the spatial prior term of the different solutions are equal, the mapping solution that is more consistent with the temporal prior term can be considered to be better.

To simultaneously optimize the data fidelity term, spatial prior term, and temporal prior term, applying an MOP to the spatiotemporal SPM problem is a good option. The three terms can be transformed into three objective functions in the MOP, as shown in (5). The optimal subpixel spatial distribution is the PS of the following:

$$\begin{aligned} \min F\left(x^c\right) &= \left[f_1\left(x^c\right), f_2\left(x^c\right), f_3\left(x^c\right)\right] \\ f_1\left(x^c\right) &= \left\|y^c - Dx^c\right\|^2 \\ f_2\left(x^c\right) &= U\left(x^c\right) = \left\|QX^c\right\|_2^2 \\ f_3\left(x^c\right) &= T\left(x^c\right) = \left\|x^c - x_h^c\right\|^2 \end{aligned} \tag{5}$$

where $f_1(x^c)$, $f_2(x^c)$, and $f_3(x^c)$ represent the data fidelity term, spatial prior term, and temporal prior term, respectively. The Laplacian prior model [42] is used as the specific spatial prior term. Q is the Laplacian matrix. The multiobjective memetic optimization method can simultaneously minimize the three prior terms, without sensitive parameters, and can achieve dynamic fusion.

B. Multiobjective Memetic Algorithm for Model Optimization

MOMSPM adopts the MOEA/D algorithm [28] as the global search procedure to explore the entire decision space, and the GSDP local search operator and MAP-based local search operator are used to further enhance the search ability and accelerate the convergence of the global population.

The flowchart of MOMSPM is illustrated in Algorithm 1.

Algorithm 1 Pseudocode of MOMSPM

```
1  Begin
2      For each class c
3          For each global search do
4              Randomly initialize a global population
                 P^c
5              Perform genetic operator x_g^c
6              Conduct local search
7                  GSDP local search x_GSDP^c
8                  MAP-based local search x_MAP^c
9              Subfitness-based population update
                 strategy P_{t+1}^c
10         End For
11     End For
12     Final HR image obtainment FHR
13     return FHR
14 End
```

Lines 2–5 and 9 describe the global search process on the basis of MOEA/D. More precisely, line 4 initializes the global population, line 5 conducts the genetic operations, and line 9 retains the elite individuals with higher subfitness from the global search and local search. The local search procedure based on GSDP local search and MAP-based local search is elaborated in lines 6–8. The local search can refine the search in a small area to improve the quality of the individual, thus improving the performance of the global search. The GSDP local search can effectively increase the spatial correlation between subpixels, and the MAP-based local search can quickly approach the optimal solution. Line 9 obtains the result map. The details are introduced as follows.

1) Population Initialization: For the convenience of calculation and evolution, an HR classification map of class c with M rows and N columns is encoded into a row vector $x^c(i)$ of size $M \times N$. The initial random population P^c of class c is made up of NP distinct individuals, i.e., $P = \{x^c(i)\}, i = 1, 2, \ldots, \mathrm{NP}$, where NP determines the diversity of the global population. The formation process is shown in the following:

$$P^c = N\left(Uy^c, \sigma^2\right) \tag{6}$$

where U is the upsampling operator of the nearest neighbor interpolation method [13]. $N(\mu, \sigma^2)$ is the Gaussian distribution, μ is its expectation, and σ^2 is its variance. Here, σ^2 is set to 0.01. The population initialization strategy is shown in Fig. 3.

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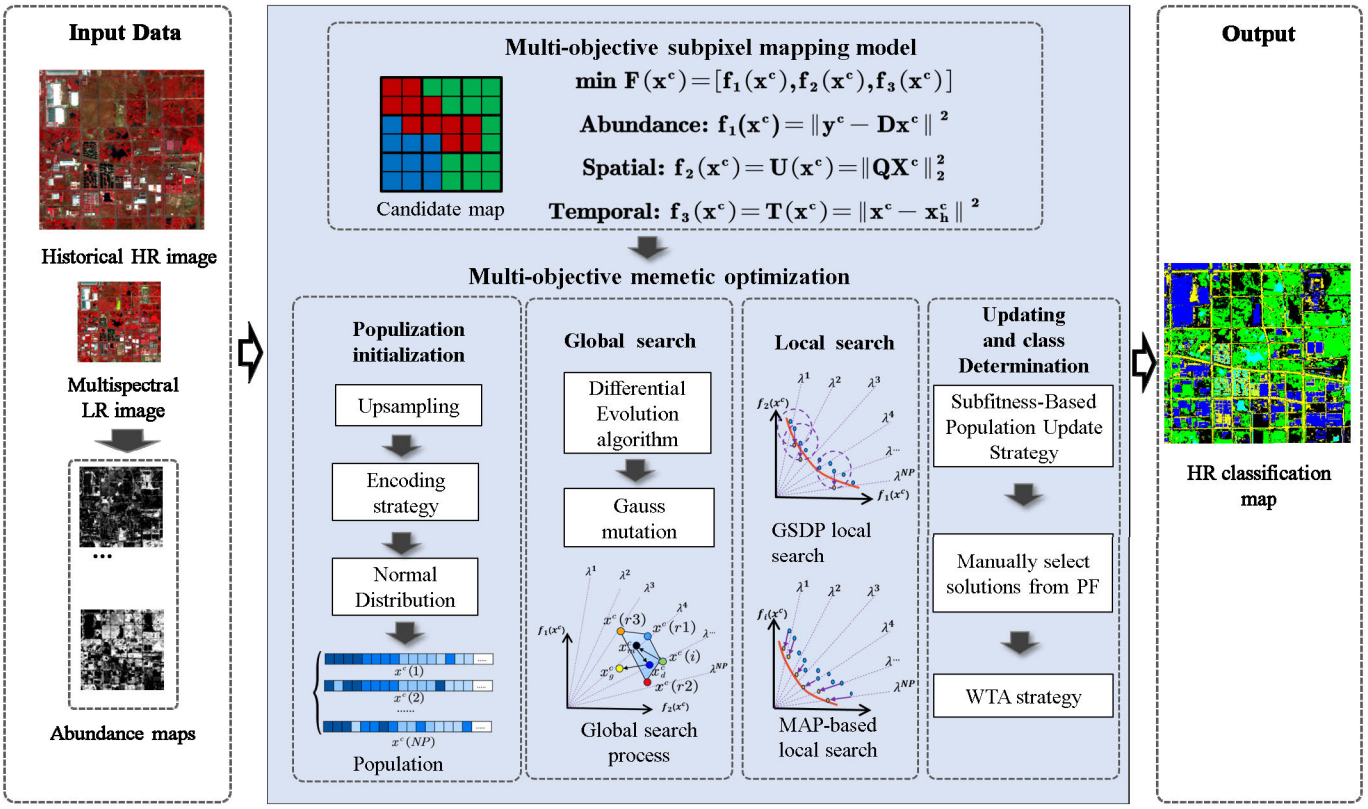


Fig. 2. Illustration of population initialization of class c.

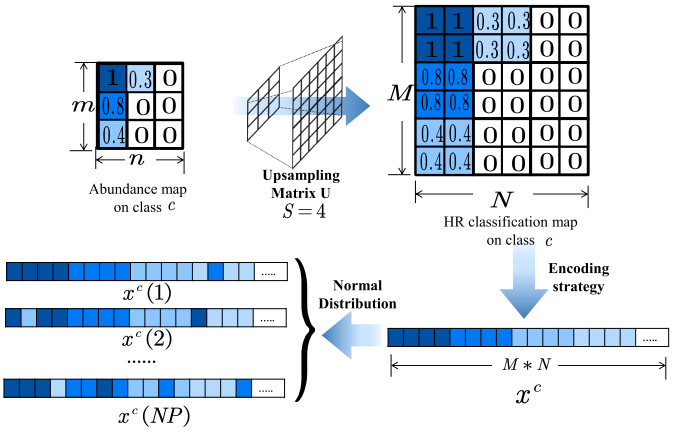


Fig. 3. Population initialization process.

MOEA/D decomposes the MOP into many different scalar subproblems. Every scalar subproblem $x^c(i)$ possesses different weight vectors $\{\lambda^1, \lambda^2, \ldots, \lambda^{\mathrm{NP}}\}$. These weight vectors $\{\lambda^1, \lambda^2, \ldots, \lambda^{\mathrm{NP}}\}$ are uniformly initialized and evenly spread in the objective space. The neighborhood of λ^i is made up of its T nearest weight vectors $C(\lambda^i) = \{\lambda^{i_1}, \lambda^{i_2}, \ldots, \lambda^{i_T}\}$. The neighborhood of x_i^c is composed of the individuals possessing the vector in $C(\lambda^i)$, i.e., $C(x_i^c) = \{x_{i_1}^c, x_{i_2}^c, \ldots, x_{i_T}^c\}$.

2) Genetic Operator: The differential evolution algorithm [45] and Gaussian mutation strategy are adopted for their powerful global search capabilities. Specifically, the DE/rand/1 strategy and the Gaussian mutation operator are formulated as shown in (7). For individual $x^c(i)$, rough searching the potential area around the solution $x^c(i)$ can generate a candidate vector x_g^c

$$
\begin{aligned}
x_m^c &= x^c(r1) + F \times \left(x^c(r2) - x^c(r3)\right) \\
x_{d,j}^c &= \begin{cases} x_{m,j}^c, & \text{if } (randb(j) \le \mathrm{CR} \text{ or } j = jrand) \\ x_j^c(i), & \text{otherwise} \end{cases} \\
x_{g,j}^c &= \begin{cases} N\left(x_{d,j}^c, \delta^2\right), & \text{if } (randb(j) < p) \\ x_{d,j}^c, & \text{otherwise} \end{cases}
\end{aligned} \tag{7}
$$

where $x^c(r1)$, $x^c(r2)$, and $x^c(r3)$ are randomly chosen and differ from $x^c(i)$. $x_j^c(i)$, $x_{m,j}^c$, $x_{d,j}^c$, and $x_{g,j}^c$ represent subpixel j in $x^c(i)$, $x_{m,j}^c$, $x_{d,j}^c$, and $x_{g,j}^c$. The scale factor is set to 0.8 and the crossover rate is set to 0.5, which can be determined by the user, and $randb(j) \in [0, 1]$ is the result of a uniform random number generator for subpixel j. $jrand \in 1, 2, \ldots, M \times N$ to ensure that x_d^c inherits at least one parameter from x_m^c. p is set to 0.1, representing the mutation probability. Fig. 4 shows the crossover mechanism for the genetic operator in MOMSPM. The different colors represent the genes (subpixels) from the different parents. The output x_g^c combines the advantages of parents with a degree of uncertainty.

3) Local Search: The goal of the local search procedure is to improve the quality of the individuals by performing a fine-grained search in a small area. Thus, MOMSPM adopts two local search strategies to improve the search capability.

The first strategy is to use the traditional pixel-swapping SPM (PSSM) algorithm [5] as the GSDP local search operator to find better solutions. Each subpixel in subproblem $x_i^c(i)$ is

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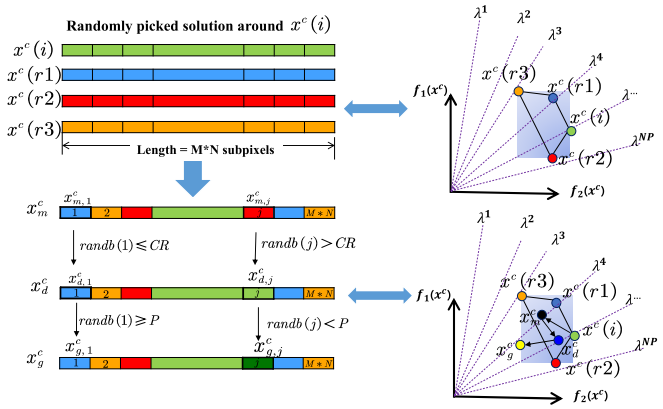


Fig. 4. Illustration of the genetic operator in MOMSPM.

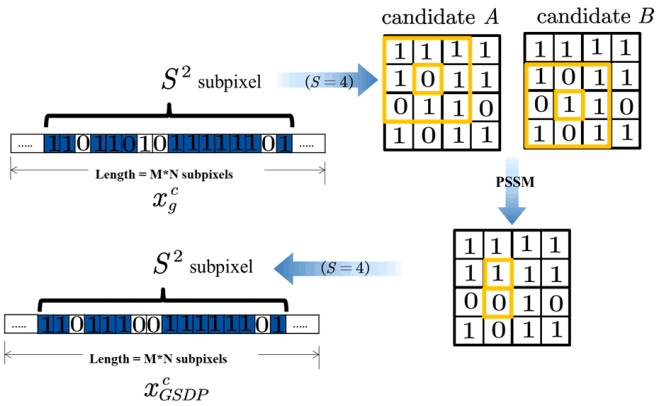


Fig. 5. Process of the GSDP local search.

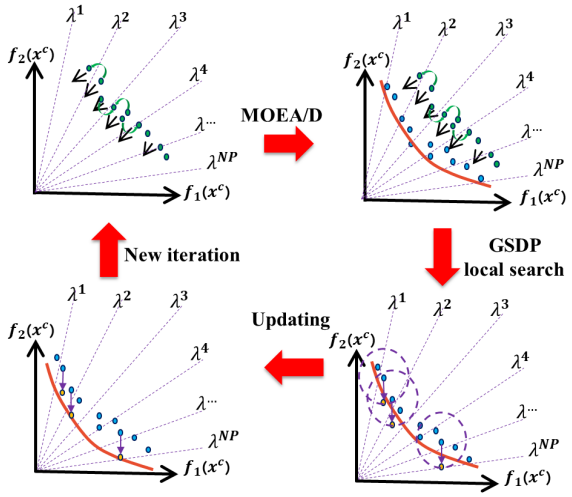


Fig. 6. Effect of the GSDP local search during population evolution.

given its neighborhood window, and the attractiveness of each subpixel can be calculated as shown in the following:

$$A_k = \sum_{P}^{p=1} \lambda_{kp} z(x_p)$$
$$\lambda_{kp} = \exp\left(\frac{-h_{kp}}{a}\right)$$
$$z(x_p) = \begin{cases} 1, & \text{if } (x_p = x_k) \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where subpixel p is the neighbor of subpixel k, and x_k and x_p are their corresponding values in $x^c(i)$. P is the overall number of subpixels in the neighborhood of x_k. $z(x_p)$ is a binary function set as 0 or 1. λ_{kp} is the distance weighting function between subpixels k and p. h_{kp} is the distance between subpixels k and p. a is the nonlinear parameter of the exponential model. For the S^2 subpixels corresponding to the mixed pixel in abundance image y^c, we find the subpixel x_k with the highest attractiveness value of 0 and the subpixel x_p with the smallest attractiveness value of 1, as shown in (9). If $A_k > A_p$, subpixels k and p are swapped

$$\begin{aligned} &\text{candidate } A = (x_k : A_k = \max(A) \mid z(x_k) = 0) \\ &\text{candidate } B = \left(x_p : A_p = \min(A) \mid z\left(x_p\right) = 1\right). \end{aligned} \tag{9}$$

The point of this step is to find the subpixels of the nontarget class mainly surrounded by the target class and the subpixels of the target class surrounded by the nontarget class. The iterations repeat until no subpixel can be exchanged and a stable SPM result is obtained. Fig. 5 shows the process of the GSDP local search in MOMSPM.

As can be seen in Fig. 5, the GSDP local search can make the subpixels of the same class appear as dense and as compact as possible in space so that the results have a higher spatial correlation. Fig. 6 shows the effect of the GSDP local search during population evolution in MOMSPM. Through swapping subpixels, the produced solution can achieve more smoothness. However, due to the principle of PSSM, the GSDP local search cannot change the abundance values. Therefore, $f_1(x^c)$ remains the same and $f_2(x^c)$ is improved.

The second option, which is the MAP-based local search operator, is to combine the weight vector of MOEA/D with the three prior terms in SST_MAP. While taking advantage of its high-speed convergence ability, MOEA/D's global optimization ability also prevents the solution set from falling into the local optimal solution. For subproblem $x^c(i)$, the objective function $f^i_{\text{MAP}}(x^c(i))$ is given as follows:

$$\begin{aligned} &f^i_{\text{MAP}}\left(x^c(i)\right) \\ &\quad = \lambda^i_1 f_1\left(x^c(i)\right) + \lambda^i_2 f_2\left(x^c(i)\right) + \lambda^i_3 f_3\left(x^c(i)\right). \end{aligned} \tag{10}$$

With regard to (10) as an energy function, its derivation is shown in the following:

$$\begin{aligned} &\nabla E_{\text{MAP}}\left(x^c\right) \\ &= -2\lambda^i_1 D^T\left(y^c - Dx^c\right) + \lambda^i_2 \nabla U\left(x^c\right) + \lambda^i_3\left(x^c - x^c_h\right). \end{aligned} \tag{11}$$

The descent iteration at epoch n can be denoted as

$$x^c_{\text{MAP}} = x^c_g + \text{lr} \cdot (\nabla E_{\text{MAP}}) \tag{12}$$

where the learning rate lr is set to 0.01.

The advantage of the MAP-based local search operator is that it can converge quickly and reduce the uncertainty in the evolutionary process. Fig. 7 shows the mechanism for the MAP-based local search in MOMSPM. $f_i(x^c)$ and $f_j(x^c)$ can represent any two different prior terms among the data fidelity term, spatial prior term, and temporal prior term.

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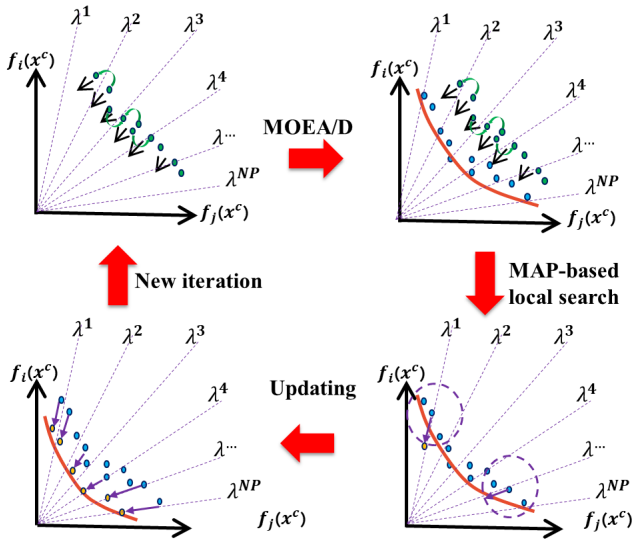


Fig. 7. Effect of the MAP-based local search during population evolution.

4) Update Strategy: The subfitness-based population update strategy [28] is adopted as the update strategy in MOMSPM for its excellent performance in generating the optimal class distribution of subpixels with different tradeoffs in multiobjective optimization.

Instead of the traditional individual update method, the update unit of the subfitness-based population update strategy is the components of the individuals. Every component is made up of S^2 subpixels corresponding to the same mixed pixel in an individual. The components with higher subfitness are supposed to survive.

The subfitness of component j in $x^c(i)$ can be calculated as follows:

$$\begin{aligned}\text{SubFit}\left(x^c(i,j)\right) &= \text{minimizeg}^{te}\left(x^c(i,j) \mid \lambda^i, z^*\right)\\ &= \max_{1\le h\le 3}\left\{\lambda_h^i \left| f_h\left(x^c(i,j)\right) - z_h^*\right|\right\}\\ &\quad \text{s.t. } x \in \Omega\end{aligned} \tag{13}$$

where $x^c(i,j)$ is the jth component in $x^c(i)$ and λ^i is the corresponding weight vector of $x^c(i)$. $z_h^* = \max\{f_h(x) \mid x \in \Omega\}$ $(h = 1, 2, 3)$ is the reference point. z_h^* can be initialized as a large value and updated during the iteration according to the real-time subfitness. $f_h(x^c(i,j))(h = 1, 2, 3)$ represents the performance of the components in the multiobjective spatiotemporal model, i.e., $f_1(x^c(i,j)) = \|y_j^c - Dx^c(i,j)\|^2$, where y_j^c is the jth pixel in y^c, $f_2(x^c(i,j)) = U(x^c(i,j))$, and $f_3(x^c(i,j)) = \|x^c(i,j) - x_h^c(i,j)\|^2$. Different weight vectors lead to various Pareto-optimal points, representing tradeoffs between $[f_1(x^c), f_2(x^c), f_3(x^c)]$. Because (5) is a minimization problem, the components with smaller subfitness are more likely to be retained. After calculating the subfitness of the components of the jth subproblem, the subfitness of their neighborhood $C(x^c(i)) = \{x^c(i_1), x^c(i_2), \ldots, x^c((i_T))\}$, and the subfitness of new individual x_g^c, the subfitness of their components is compared. The component with higher subfitness can survive $x_{t+1}^c(i)$. When the GSDP local search strategy is adopted, the comparisons are given as follows:

$$\begin{aligned}&x_{\text{new},j}^c\\ &= \begin{cases} x_{m,j}^c, & \text{if SubFit}\left(x_{m,j}^c\right) < \text{SubFit}\left(x_{\text{GSDP},j}^c\right)\\ x_{\text{GSDP},j}^c, & \text{if SubFit}\left(x_{m,j}^c\right)\ \text{SubFit}\left(x_{\text{GSDP},j}^c\right)\end{cases}\\ &x_{t+1}^c(i,j)\\ &= \begin{cases} x_{\text{new},j}^c, & \text{if SubFit}\left(x_{\text{new},j}^c\right) < \text{SubFit}\left(C\left(x^c(i,j)\right)\right)\\ C\left(x^c(i,j)\right), & \text{otherwise.}\end{cases}\end{aligned} \tag{14}$$

When the MAP-based strategy is conducted, the comparisons are given as follows:

$$\begin{aligned}&x_{\text{new},j}^c\\ &= \begin{cases} x_{m,j}^c, & \text{if SubFit}\left(x_{m,j}^c\right) < \text{SubFit}\left(x_{\text{MAP},j}^c\right)\\ x_{\text{MAP},j}^c, & \text{if SubFit}\left(x_{m,j}^c\right)\ \text{SubFit}\left(x_{\text{MAP},j}^c\right)\end{cases}\\ &x_{t+1}^c(i,j)\\ &= \begin{cases} x_{\text{new},j}^c, & \text{if SubFit}\left(x_{\text{new},j}^c\right) < \text{SubFit}\left(C\left(x^c(i,j)\right)\right)\\ C\left(x^c(i,j)\right), & \text{otherwise.}\end{cases}\end{aligned} \tag{15}$$

For each individual in the old population P_t^c, the process of reproduction and population update is repeated until one generation is completed, forming a new population with NP better solutions, i.e., $P_{t+1}^c = \{x_{t+1}^c(i)\}, i = 1, 2, \ldots, \text{NP}$. When the number of iterations reaches the stopping condition, the final result is the generated PF.

5) Final HR Image Obtainment: After the evolution process is completed, for each class c, the user can then manually choose the best solutions x_*^c from the PF, based on their preferences. Thus, a set of class maps $x = [x_*^1, x_*^2, \ldots, x_*^C]$ is obtained. The class maps in x are combined to create the final HR classification map FHR via the winner-takes-all (WTA) class determination strategy [46], as defined in the following equation, where $j = 1, 2, \ldots, M \times N$

$$\text{FHR} = \arg\max\left\{x_{*,j}^c \mid c = 1, 2, \ldots, C\right\}. \tag{16}$$

IV. Experiments and Analysis

A. Experimental Settings

1) Data Preparation: In order to evaluate the performance of the proposed method, one simulated image and two real images were used in the experiments. The simulated abundance images were acquired by downsampling the reference classification image, which can reduce the uncertainty caused by errors from the sensor PSF, atmospheric and geometric effects, co-registration operation, and spectral unmixing. In contrast, the pairs of HR and LR images used in the real-data experiments were acquired by sensors in a real environment from the same region and phase. The real-data experiments can evaluate the adaptability and tolerability of the various SPM algorithms to the uncertainty.

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2) Comparison Methods: In order to validate the effectiveness of MOMSPM, seven other methods were chosen as comparative SPM algorithms. Four traditional SPM algorithms [SASM (Mertens et al. [7]), PSSM (Atkinson [5]), AMCDSM_L (Zhong et al. [46]), and MOSM (Ma et al. [28])] and three spatiotemporal SPM methods [SLCCM (Ling et al. [15]), SSM_CS (He et al. [16]), and SST_MAP (He et al. [21])] were selected. The proposed method with GSDP local search and MAP-based local search is denoted as MOMSPM-GSDP and MOMSPM-MAP, respectively.

In this article, the overall accuracy (OA) and the kappa coefficient (Cohen, 1960) are adopted for the accuracy assessment. The size of the population and the size of the neighboring individuals were set to 50 and 8, respectively. The maximum number of iterations was 30.

B. Experiment I: Landsat 8 Images (Cili)

For the synthetic Cili dataset, the original multispectral images [see Fig. 8(a)] were obtained by the Operational Land Imager (OLI) of the Landsat 8 satellite on August 7, 2013, in Cili, Hunan, China. The images are made up of six bands, 240 lines, and 240 columns, with five categories of water, farmland, vegetation, urban, and bare soil. The support vector machine (SVM) classification algorithm was employed to produce an accurate classification map as the reference map [see Fig. 8(b)] for SPM. The SVM classification for the Landsat 8 images captured on July 13 and August 7 achieved high OA values (90.36% and 93.91%, respectively), demonstrating the reliability of the ground truth used for validating the SPM algorithms. The 30-m spatial resolution reference map was then downsampled to create the 120-m spatial resolution synthetic abundance maps [see Fig. 8(e)] with a scale factor of 4.

While the SPM methods reconstructed the fine-resolution reference map only with the LR abundance maps, the spatiotemporal SPM methods utilized additional historical HR imagery. For the historical Landsat 8 image acquired on July 13, 2013, SVM was also applied to generate the classification map [see Fig. 8(d)], the spatial resolution of which was also 30 m. July–August is the harvest season in the Cili area when a large number of farmland crops are matured and harvested, so that a certain proportion of the farmland turns into bare soil. As for the other classes of water, vegetation, and urban, the few changes in these classes indicate that the temporal information can be of great use. The overall land-cover change ratio between the historical classification map and reference map [see Fig. 8(b) and (d)] is 20.54%.

The mapping results for the Cili dataset are shown in Fig. 8(f)–(n). To observe the detailed restoration pattern, zoomed-in view areas are shown in Fig. 9(a)–(k). The corresponding quantitative accuracies are listed in Table I. A visual comparison of the results shows that the minimization-based spatiotemporal SPM methods (SLCCM, SSM_CS, SST_MAP, MOMSPM-GSDP, and MOMSPM-MAP) are preferable to the traditional SPM methods (SASM, PSSM, AMCDSM, and MOSM). The results of SASM and PSSM show slight artifacts, and the results of AMCDSM and MOSM are oversmoothed by their spatial regularization term. It is unfortunate that all of the traditional SPM methods listed above struggle with preserving linear features and handling complex distribution patterns, such as the linear urban area in the farmland, the zoomed-in view areas of the Cili dataset, farmland, bare soil, and the mixed vegetation areas. Besides, the spatial distribution in the Cili dataset lacks regular shapes and fixed distribution patterns, which is very challenging for the different SPM algorithms. Compared to the traditional SPM methods, the spatiotemporal SPM methods show great potential to leverage temporal and spatial correlations, and among these methods, MOMSPM-GSDP and MOMSPM-MAP have proven to be the most effective in achieving satisfactory performance.

C. Experiment II: GF-2 and Landsat 8 Images

To validate the ability of the various algorithms applied in a practical situation, an experiment with 4-m bitemporal Gaofen-2 (GF-2) satellite multispectral sensor (MSS) images from Wuhan, Hubei, China, was conducted. The original bitemporal GF-2 images were acquired on April 22, 2015 [see Fig. 10(a)], and April 21, 2016 [see Fig. 10(c)]. The SVM classification algorithm was applied to the bitemporal GF-2 images to produce a reliable classification map as the historical thematic map [see Fig. 10(b)] and the reference map [see Fig. 10(d)]. The SVM classification for the GF-2 images from 2015 and 2016 achieved high OA values (97.31% and 96.79%, respectively), demonstrating the reliability of the ground truth used for validating the SPM algorithms. Due to the rapid urbanization of Wuhan, the change ratio between the bitemporal classification maps is 32.56%, and the change of each class is not small. Large numbers of subpixels of vegetation and water change and blend. A considerable percentage of bare soil is present, and the shape of the bare soil areas changes dramatically.

As for the input real LR image, the 30-m multispectral image [see Fig. 10(e)] was acquired by the Landsat 8 OLI on March 2, 2016, covering the same area as the original bitemporal GF-2 images. The selected multispectral image was made up of seven bands, 160 lines, and 160 columns, and the historical thematic map and the reference map consisted of 960 lines, 960 columns, and five classes of water, vegetation, bare soil, building, and road. Therefore, the mapping scale factor S was set to 6 and the original 4-m GF-2 images were resized to 5-m images to match the 30-m Landsat 8 image by six times.

The time-series imagery acquired at different times needed to be transformed into the same coordinate system to maintain consistency. With the help of the co-registration toolbox in ENVI, it was ensured that the coordinates of the predicted subpixel in the Landsat 8 image and the fine pixel in the GF-2 image were the same. The root-mean-square error (RMSE) of the co-registration was 0.49.

We manually selected five classes existing in the reference map as the endmembers. The fully constrained least squares (FCLS) algorithm [37] was then used to decompose the

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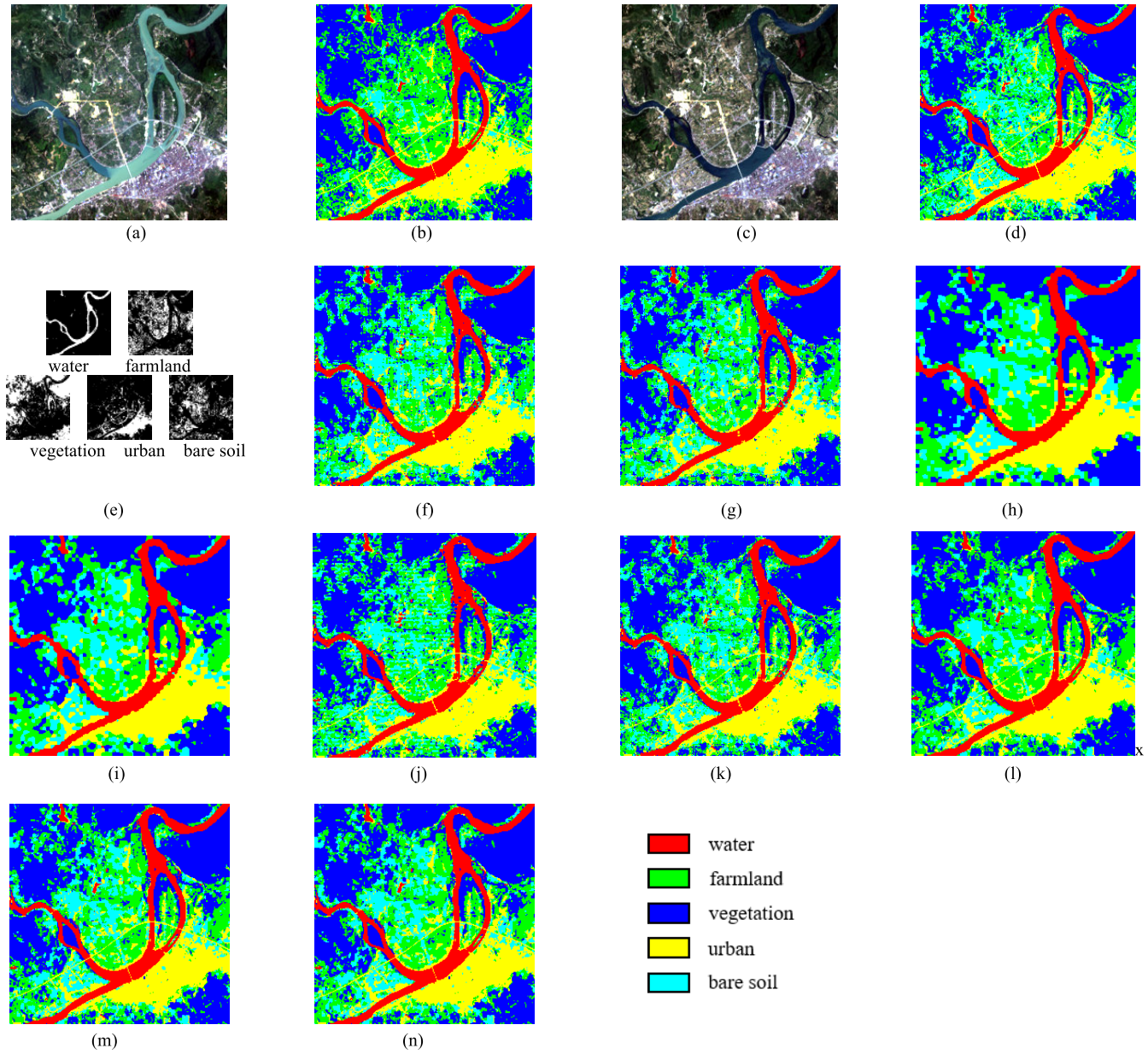


Fig. 8. SPM results for Experiment I. (a) Landsat 8 image on July 13, 2013. (b) SVM classification result for the July 13, 2013, image. (c) Landsat 8 image on August 7, 2013. (d) SVM classification result for the August 7, 2013, image. (e) Abundance images. (f) SASM. (g) PSSM. (h) AMCDSM. (i) MOSM. (j) SLCCM. (k) SSM_CS. (l) SST_MAP. (m) MOMSPM-GSDP. (n) MOMSPM-MAP.

TABLE I
MAPPING ACCURACIES FOR EXPERIMENT I

	SASM	PSSM	AMCDSM	MOSM	SLCCM	SSM_CS	SST_MAP	**MOMSPM-GSDP**	**MOMSPM-MAP**
Water	94.70	95.41	95.32	96.17	96.86	97.34	98.96	99.17	99.10
Farmland	58.48	58.16	62.81	63.55	69.47	68.41	79.10	76.08	76.02
Vegetation	84.95	84.19	87.26	87.60	87.64	89.99	91.80	92.03	91.87
Urban	77.54	77.26	76.47	78.97	84.17	88.21	89.89	90.96	90.74
Bare soil	57.13	55.69	57.33	60.89	63.46	65.75	53.30	**66.30**	**66.84**
OA (%)	74.26	73.71	76.01	77.34	79.98	81.61	83.12	**84.79**	**84.77**
Kappa	0.6618	0.6546	0.6839	0.7018	0.7375	0.7584	0.7770	**0.7998**	**0.7995**

spectral signature of the LR Landsat 8 images and generate the corresponding abundance maps.

The SPM of the Wuhan dataset is rather challenging, not only because of the unmixing error and registration error but also because of the complex spatial distributions contained in this HR (4 m) dataset. The results of the various SPM algorithms are given in Fig. 10(g)–(o). Fig. 11(a)–(k) shows the corresponding zoomed-in view regions for closer observation

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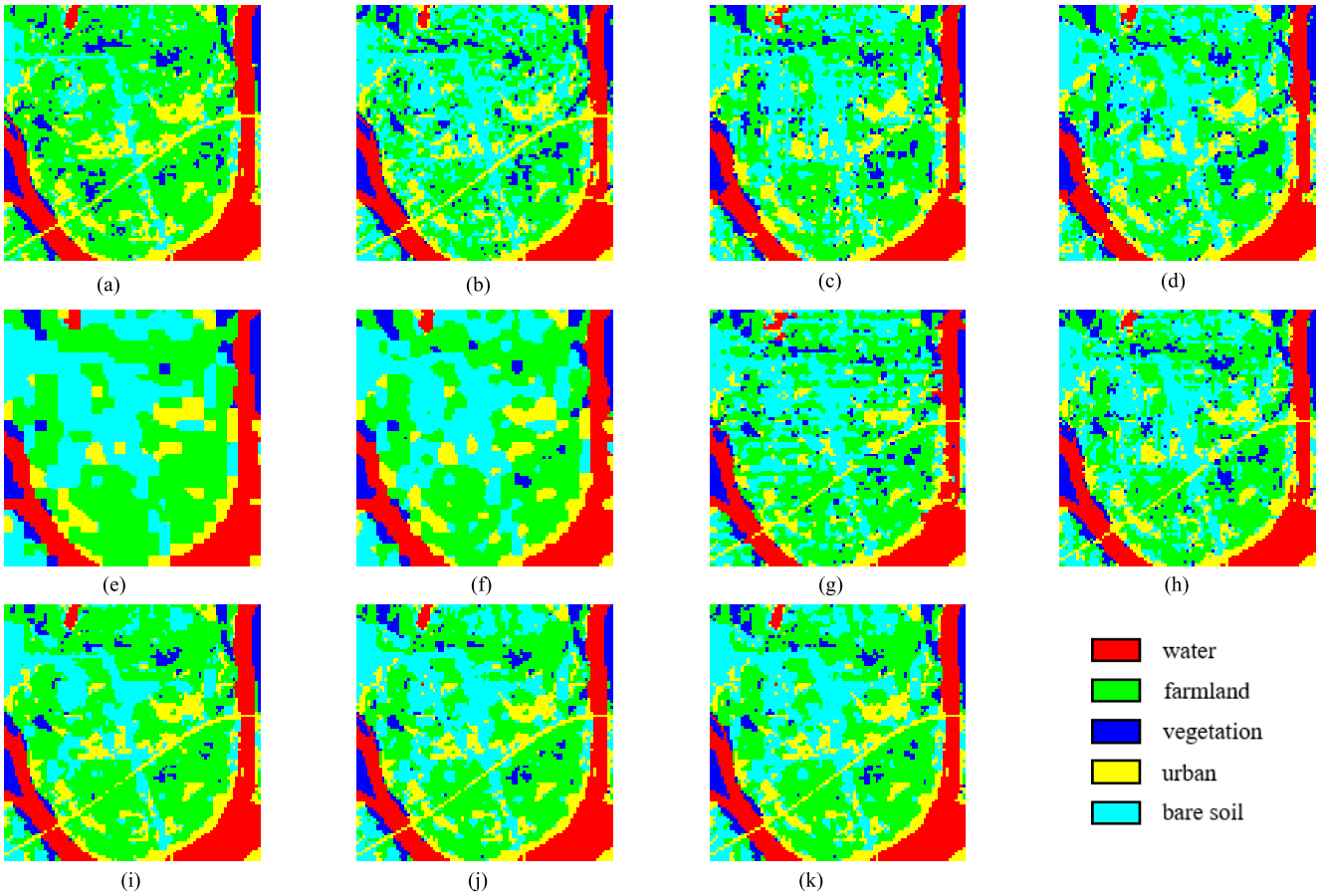


Fig. 9. Detailed regions of the SPM results for Experiment I. (a) Zoomed-in view of the historical HR image from July 13, 2013. (b) Zoomed-in view of the reference image from August 7, 2013. (c) SASM. (d) PSSM. (e) AMCDSM. (f) MOSM. (g) SLCCM. (h) SSM_CS. (i) SST_MAP. (j) MOMSPM-GSDP. (k) MOMSPM-MAP.

of the detailed mapping results. The quantitative assessment results for the OA, kappa, and producer's accuracy are listed in Table II.

From the results shown in Figs. 10 and 11, artifacts and jagged boundaries are obvious in the result maps of SASM and PSSM, as a result of the misleading spectral unmixing error. The other traditional SPM methods, such as AMCDSM and MOSM, produce a smoother spatial distribution for the spatial prior, suppressing the uncertainty brought by spectral unmixing. However, AMCDSM and MOSM are short of detailed structural information, especially for the bare soil and building classes. Despite exploiting the spatiotemporal information from the time-series images, SLCCM and SSMCS perform poorly in this real-data experiment, considering their performance in the synthetic image experiment. The ubiquitous salt-and-pepper noise found in the result maps of SLCCM and SSMCS is strong evidence that they are affected by the spectral unmixing error. As for SST_MAP, it is more robust to errors in the abundance maps and reconstructs contiguous land-cover boundaries for the buildings, roads, and bare soil areas. However, compared with MOMSPM-GSDP and MOMSPM-MAP, the results of SST_MAP are too smooth to predict the fine distribution details of buildings and water in particular regions.

The integration of MOEA/D in the proposed methods can better balance the influence of spatiotemporal information and abundance information, thereby producing more accurate class distribution details. It can not only effectively retain the unaltered road network but can also accurately classify the new buildings. Furthermore, the local operator can fine-tune the details of the ground objects to obtain a dense and compact effect.

From the statistical results listed in Table II, it can be seen that the proposed MOMSPM-GSDP model improves the accuracy for water and building by 4.76% and 8.66%, respectively, reaching 68.34% and 38.21%. The proposed MOMSPM-MAP model improves the accuracy for water and building by 5.00% and 9.86%, respectively, reaching 68.58% and 39.41%. With respect to the OA and kappa, the proposed methods achieve the best results, exceeding the SST_MAP model by 1.5% in OA.

D. Experiment III: Sentinel-2A and Landsat 8 Images

To further inspect the capability of the proposed methods, we constructed a real-data experiment with 10-m bitemporal Sentinel-2A images and a 30-m Landsat 8 image. The bitemporal images were acquired on November 13, 2016 [see Fig. 12(a)], and October 9, 2017 [see Fig. 12(c)], covering Nanjing, Jiangsu, China. The Landsat 8 image acquired on October 9, 2017 [see Fig. 12(e)], was adopted to generate the abundance maps by FCLS spectral unmixing. The five endmembers of vegetation, building, bare soil, water, and

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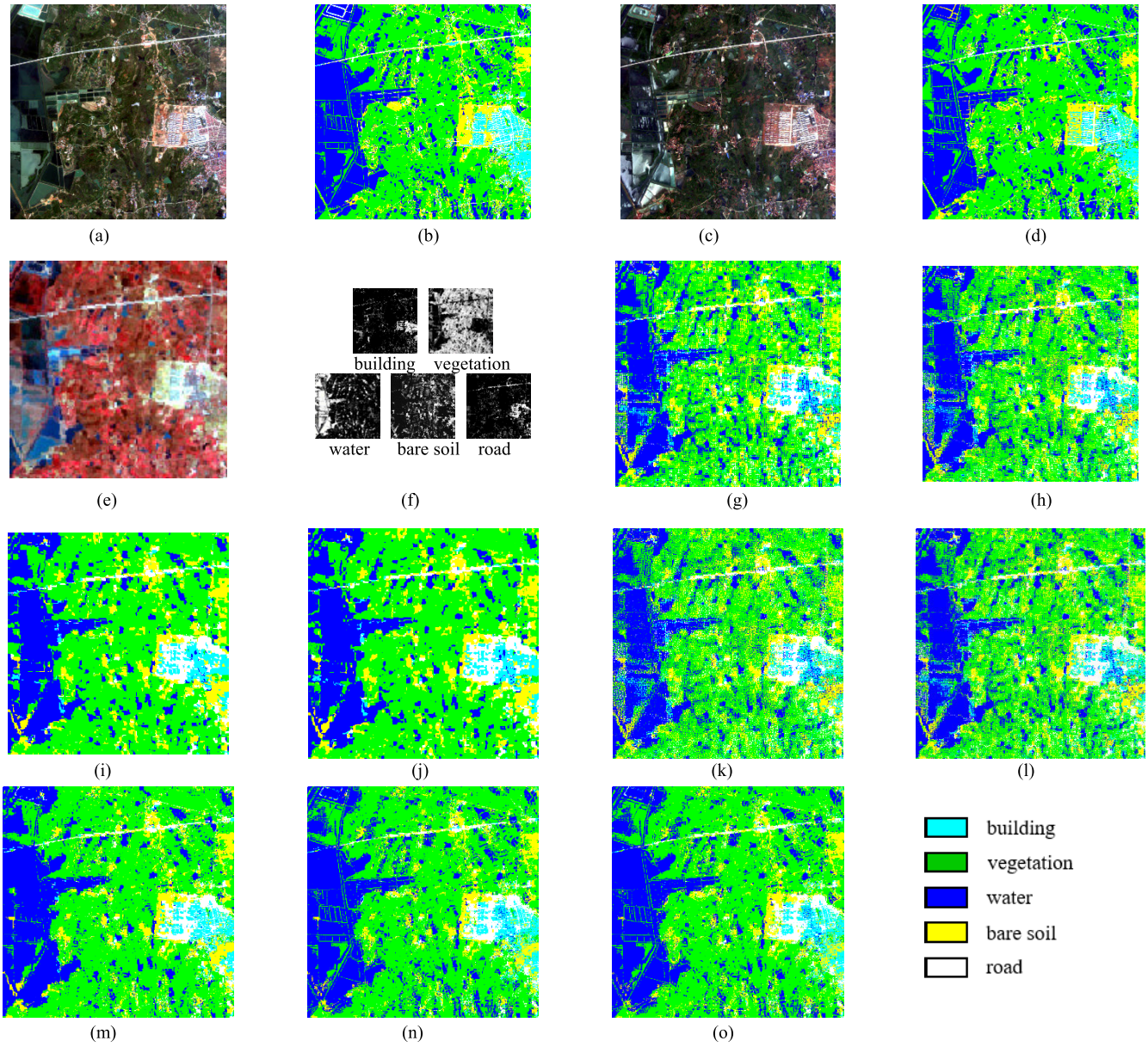


Fig. 10. SPM results for the GF-2 and Landsat 8 images in Experiment II. (a) Original GF-2 image from 2015. (b) SVM classification result for the 2015 image. (c) Original GF-2 image from 2016. (d) SVM classification result for the 2016 image. (e) Original Landsat 8 image from 2016. (f) Abundance images. (g) SASM. (h) PSSM. (i) AMCDSM. (j) MOSM. (k) SLCCM. (l) SSMCS. (m) SST_MAP. (n) MOMSPM-GSDP. (o) MOMSPM-MAP.

TABLE II
MAPPING ACCURACIES FOR EXPERIMENT II

	SASM	PSSM	AMCDSM	MOSM	SLCCM	SSM_CS	SST_MAP	**MOMSPM-GSDP**	**MOMSPM-MAP**
Building	27.11	26.97	30.22	30.17	26.67	26.69	29.55	**38.21**	**39.41**
Vegetation	60.71	60.64	74.64	74.48	60.86	60.91	76.23	77.04	76.42
Water	53.08	52.97	56.15	55.89	52.15	53.31	63.58	**68.34**	**68.58**
Bare soil	28.61	28.69	31.59	31.48	28.38	28.75	33.41	26.90	29.59
Road	31.50	31.19	33.71	33.19	31.04	31.37	33.53	30.36	29.70
OA (%)	53.33	53.25	62.55	62.36	53.12	53.50	65.35	**66.90**	**66.90**
Kappa	0.2749	0.2737	0.3762	0.3733	0.2717	0.2776	0.4212	**0.4460**	**0.4478**

road were manually selected and the abundance maps were obtained, as shown in Fig. 12(f).

The classification maps generated from the bitemporal images by SVM were adopted as the historical thematic map and reference map. The SVM classification for the Sentinel-2A images from November 13, 2016, and October 9, 2017, achieved high OA values (95.57 and 94.46%, respectively), demonstrating the reliability of the ground truth used for

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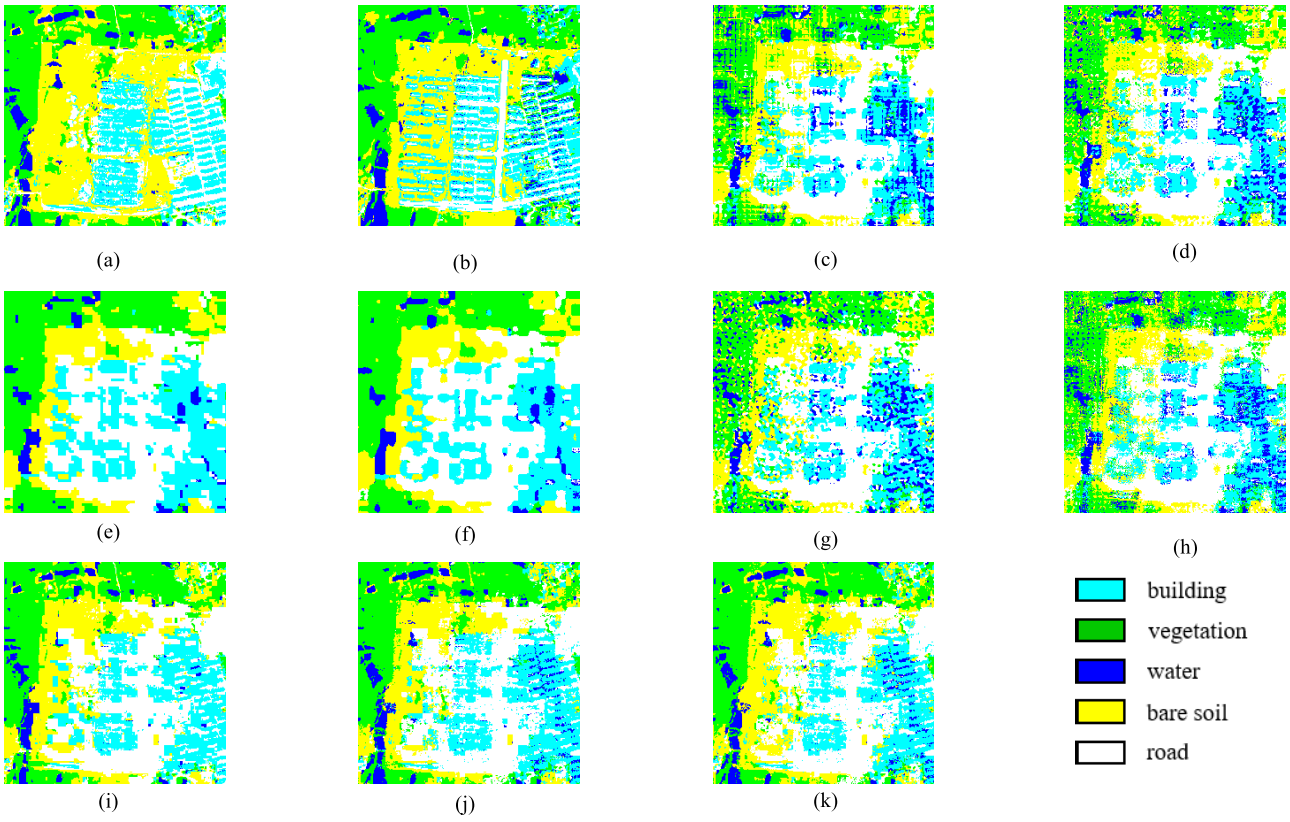


Fig. 11. Detailed regions of the SPM results for the GF-2 and Landsat 8 images in Experiment II. (a) Zoomed-in view of the historical HR image from July 13, 2013. (b) Zoomed-in view of the reference image from August 7, 2013. (c) SASM. (d) PSSM. (e) AMCDSM. (f) MOSM. (g) SLCCM. (h) SSMCS. (i) SST_MAP. (j) MOMSPM-GSDP. (k) MOMSPM-MAP.

validating the SPM algorithms. The change ratio between the historical thematic map and the reference map is 36.18%, which is mainly caused by new buildings, increased vegetation, and reduced bare soil. Meanwhile, the backbone of the road network remains unchanged, but some minor changes have taken place in the branches. The co-registration operation was again necessary, for which the RMSE was 0.31. The spatial resolution of the abundance maps and the reference map was 30 and 10 m, respectively, with the size of $140 \times 140 \times 7$ pixels and $420 \times 420 \times 4$ pixels, respectively. Therefore, the predefined scale factor S was 4.

The uncertainty in the abundance map, co-registration error, and misleading temporal information brought by the high change ratio between bitemporal images fully tested the robustness and effectiveness of the different SPM methods. The mapping results for the Nanjing dataset are shown in Figs. 12 and 13.

From the global and detailed views shown in Figs. 12 and 13, the mapping results of SASM, PSSM, SLCCM, and SSM_CS are affected by the unmixing error, and thus, speckle noise is common. Many subpixels that belong to bare soil are misclassified as vegetation and buildings. In comparison, AMCDSM and MOSM can reduce the impact of the unmixing error with the spatial prior term, but they fail to reconstruct the shape of the bare soil and the slender road due to the limited abundance information. Differing from the aforementioned SPM methods, the road network in the results of SST_MAP, MOMSPM-GSDP, and MOMSPM-MAP preserves the smoothness and continuity, as a result of their excellent ability to balance the abundance constraint, spatial dependence, and temporal dependence. However, as for the restoration details of roads, MOMSPM-GSDP and MOMSPM-MAP achieve better performances than SST_MAP.

The mapping accuracies for the Nanjing dataset are presented in Table III. From Table III, the significant improvement in the OA of water and road indicates the superior performance of the multiobjective memetic algorithm in dealing with the SPM problem.

E. Optimization Analysis

To further demonstrate the effectiveness of the multiobjective memetic algorithm, we took the Cili dataset as an example. The optimization results of a variant algorithm without local search (MOMSM-noLocal) were compared with those of MOMSPM-GSDP and MOMSPM-MAP. Their PFs for the bare soil class are shown in Fig. 14. Since the multiobjective spatiotemporal SPM model is a problem of minimization (13), the population near the origin is better. As can be seen in Fig. 14, the PFs are uniformly distributed, and any slight improvement in one objective can lead to degradation in another objective. It is worth noting that the PF of MOMSPM-MAP has wider coverage than that of MOMSPM-GSDP and MOMSPM-MAP, as its local operator can easily find the optimal solutions in different directions.

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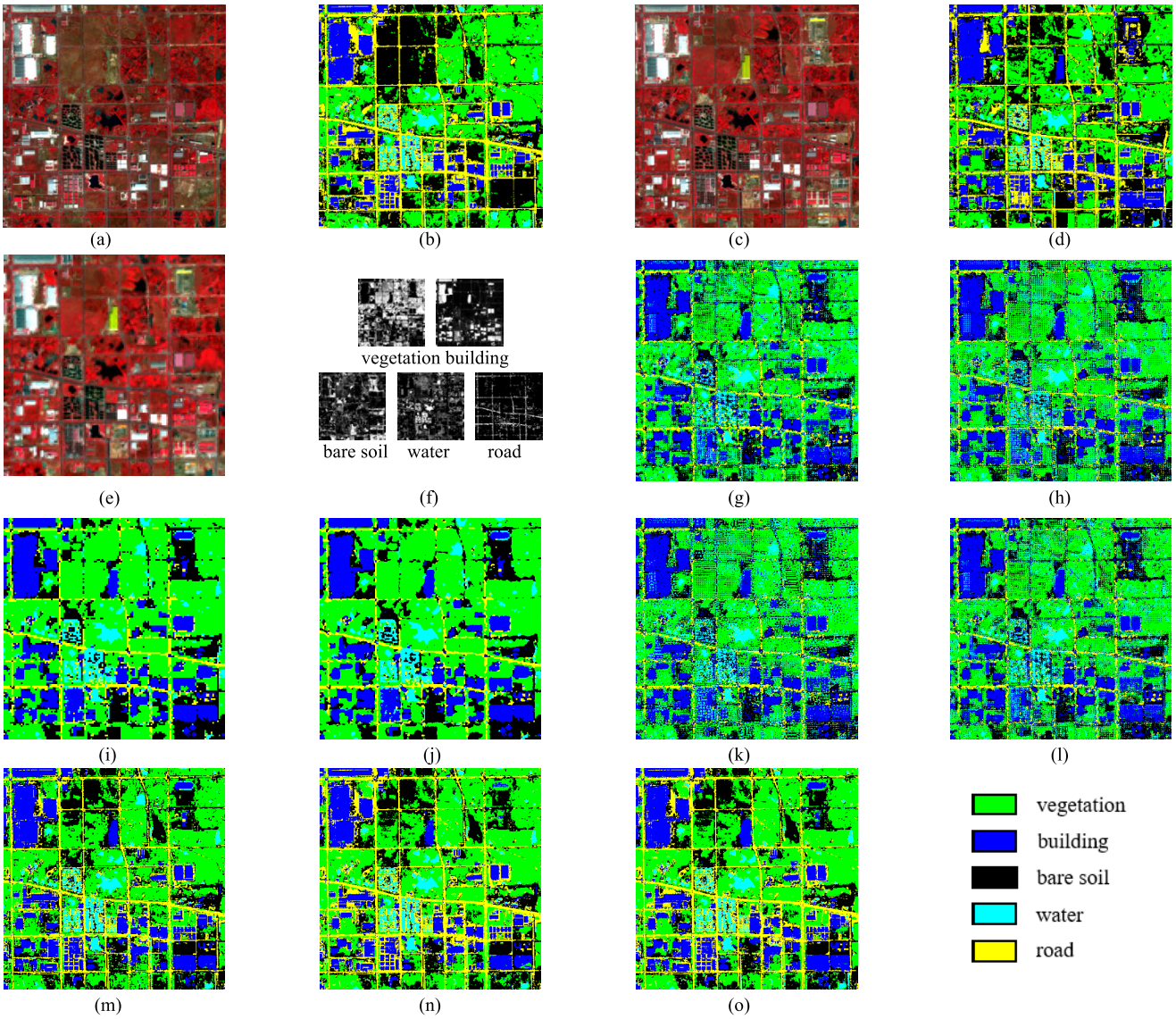


Fig. 12. SPM results for the Sentinel-2A and Landsat 8 Images in Experiment III.(a) Original Sentinel-2A image from 2016. (b) SVM classification result for the 2016 image. (c) Original Sentinel-2A image from 2017. (d) SVM classification result for the 2017 image. (e) Original Landsat 8 image from 2017. (f) Abundance images. (g) SASM. (h) PSSM. (i) AMCDSM. (j) MOSM. (k) SLCCM. (l) SSMCS. (m) SST_MAP. (n) MOMSPM-GSDP. (o) MOMSPM-MAP.

TABLE III
MAPPING ACCURACIES FOR EXPERIMENT III

	SASM	PSSM	AMCDSM	MOSM	SLCCM	SSM_CS	SST_MAP	MOMSPM-GSDP	MOMSPM-MAP
Vegetation	69.54	69.45	88.30	88.66	72.20	70.98	81.58	**84.74**	**84.74**
Building	68.66	67.59	74.42	75.54	69.57	70.22	75.92	71.86	71.86
Bare soil	35.75	34.90	34.82	34.82	35.93	36.81	54.87	49.77	49.77
Water	56.70	54.90	66.34	66.86	62.47	60.82	82.22	79.59	79.59
Road	15.90	16.03	19.55	19.03	15.66	17.13	55.83	**69.12**	**69.12**
OA (%)	48.98	48.49	56.62	57.04	50.18	50.38	67.97	**69.19**	**69.19**
Kappa	0.3323	0.3259	0.4139	0.4196	0.3469	0.3496	0.5707	**0.5881**	**0.5881**

We also measured the performance of the multiobjective memetic algorithm by hypervolume (HV) [47] and pure diversity (PD) [48]. HV measures the size of the region dominated by the nondominated solution set, and PD evaluates the diversity of the population. The reference point of HV was (5000, 150 000, 50 000). Tables IV and V list the final HV and PD values for the different classes when processing the Cili data. Due to the principles of the different algorithms, there is a slight difference in the HV and PD between MOMSPM-GSDP and MOMSPM-noLocal, but MOMSPM-MAP shows an obvious increase in HV and PD. The increase in OA and kappa also indicates that the proposed method yields better solutions. The OA and kappa of MOMSM-noLocal are 84.64% and 0.7976, respectively, which are lower than those of MOMSPM-GSDP

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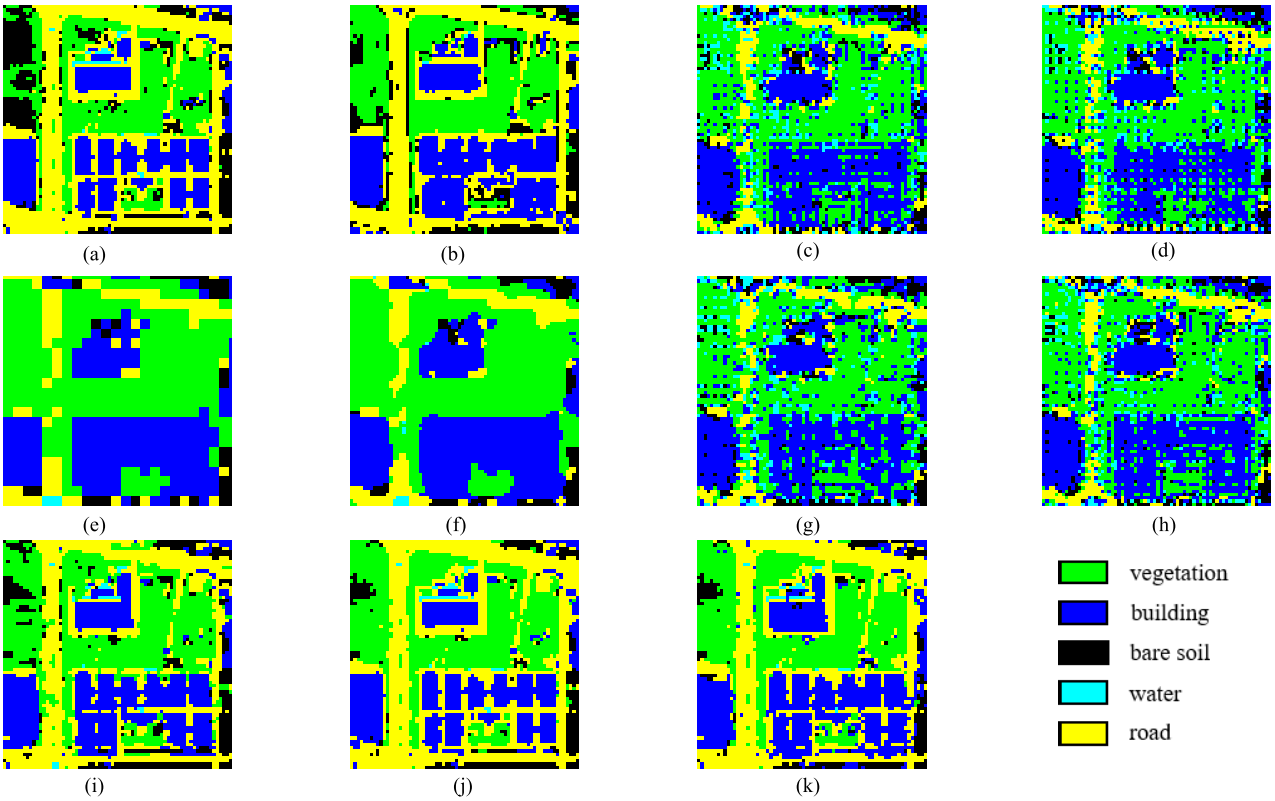


Fig. 13. Detailed regions of the SPM results for the Sentinel-2A and Landsat 8 Images in Experiment III. (a) Zoomed-in view of the historical HR image from 2016. (b) Zoomed-in view of the reference image from 2017. (c) SASM. (d) PSSM. (e) AMCDSM. (f) MOSM. (g) SLCCM. (h) SSMCS. (i) SST_MAP. (j) MOMSPM-GSDP. (k) MOMSPM-MAP.

(84.79% and 0.7998) and MOMSPM-MAP (84.77% and 0.7995).

In addition, the stopping condition of the proposed method was set to 30 because the high-dimensional solution is divided into multiple low-dimensional mapping blocks (the components of the individuals) for optimization, which allows for convergence in only a few iterations. The experimental results also indicate that the population of the mapping solutions stabilizes after 25 iterations. To illustrate the rationality of this stopping criterion, we employed HV to measure the quality of the population convergence after each iteration of MOMSPM-GSDP and MOMSPM-MAP. Fig. 15 shows the changes in HV values during the iterations when applying the MOMSPM-GSDP method. The increasing HV with iterations implies that the population generated more accurate mapping solutions during evolution. However, after 25 iterations, the improvement of HV gradually levels off, indicating that the population has reached convergence. Setting a larger number of iterations as the stopping condition can ensure population convergence, but this will also increase the computational time cost.

F. Complexity Analysis

The complexity of MOMSPM includes three parts: 1) population initialization; 2) global and local search; and 3) final HR image obtainment. NP is the size of the population, and M and N are the row number and column number in the HR image, respectively. C is the number of different classes. G represents the total number of generations. W is the iteration of the MAP-based local operator or GSDP local operator.

1) The time complexity of the population initialization is $O(\text{NP} \times M \times N \times C)$. The complexity of the global search, including reproduction and the subfitness-based population update, is $O(\text{NP} \times M \times N \times C)$.
2) The complexity of the local search is $O(\text{NP} \times M \times N \times W \times G \times C)$.
3) The complexity of the final HR image obtainment process is $O(\text{NP} \times C)$.

To further illustrate this issue, we measured the average computation time of each method on the Cili, Wuhan, and Nanjing datasets. The results are provided in Table VI. All the experiments were conducted on a personal computer equipped with a 2.7-GHz CPU and 8 GB of RAM, using MATLAB R2016a. The results of the experiments indicate that SASM, PSSM, and SLCCM are the most efficient algorithms, due to their simple rules without the need for model optimization. The processing time of the MAP-based algorithms (AMCDSM and SST_MAP) is acceptable for small images but rapidly increases with larger image sizes. SSM CS is less time-consuming because of its pixel-wise optimization mechanism. In contrast, the MOEA-based methods (MOSM, MOMSPM-GSDP, and MOMSPM-MAP) are lower in efficiency, especially for the larger Nanjing and Wuhan datasets. The proposed methods—MOMSPM-GSDP and MOMSPM-MAP—are more time-consuming than MOSM. This is to be expected as the proposed methods use a combination of global search and local search to find the optimal solutions. However, it has also been shown that the proposed methods can achieve

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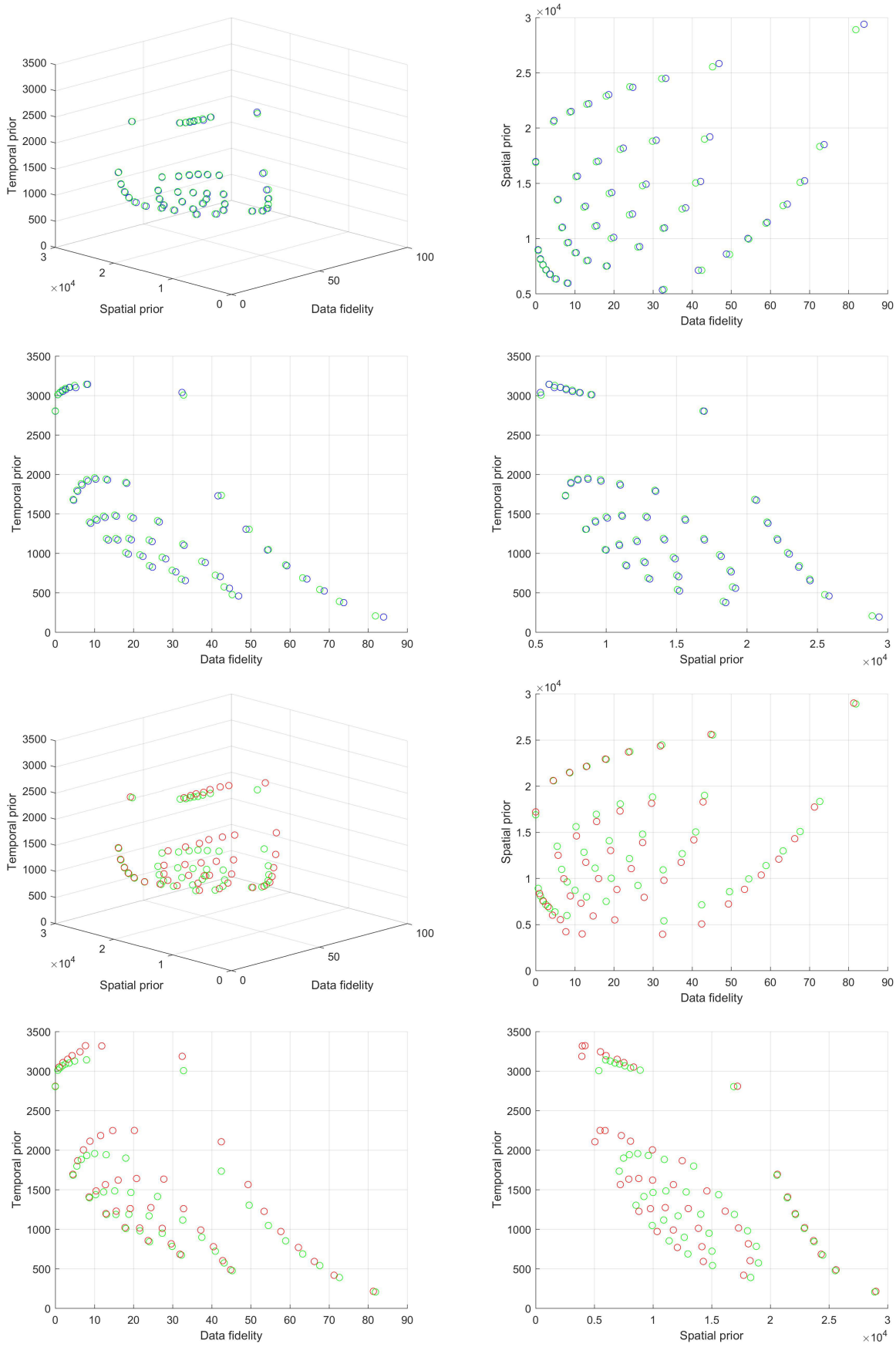


Fig. 14. Pareto fronts for the bare soil class in the Cili dataset.

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TABLE IV
HV FOR THE CILI DATASET

	Water	Farmland	Vegetation	Urban	Bare soil
MOMSPM-noLocal	0.9928	0.9411	0.9633	0.9696	0.9586
MOMSPM-GSDP	0.9933	0.9408	0.9635	0.9696	0.9581
MOMSPM-MAP	0.9942	0.9506	0.9704	0.9781	0.9675

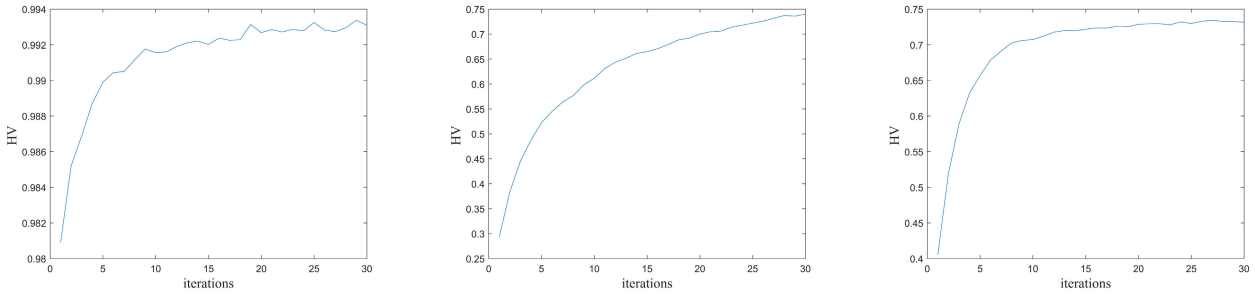


Fig. 15. HV of the population during optimization. (a) Cili, water. (b) Wuhan, building. (c) Nanjing, bare soil.

TABLE V
PD FOR THE CILI DATASET

	Water	Farmland	Vegetation	Urban	Bare soil
MOMSPM-noLocal	3.7496E7	5.3041E8	2.9905E8	1.3594 E8	2.6838E8
MOMSPM-GSDP	4.0054E7	5.5768E8	2.8693E8	1.3214E8	2.6952E8
MOMSPM-MAP	4.1726E7	6.2192 E8	3.4455E8	1.5358E8	2.8503E8

TABLE VI
COMPUTATION TIMES (S) OF THE SPM METHODS IN EXPERIMENTS I–III

	SASM	PSSM	AMCDSM	MOSM	SLCCM	SSM_CS	SST_MAP	**MOMSPM-GSDP**	**MOMSPM-MAP**
Cili	0.43	1.61	8.28	854.65	1.97	96.97	66.01	15039.98	15306.21
Wuhan	6.09	51.05	155.86	5806.53	148.44	1615.17	26410.83	142467.97	148335.71
Nanjing	1.75	5.17	22.17	3938.14	8.57	685.51	1641.53	71095.51	86639.15

better SPM results than some of the other methods, as demonstrated in Sections IV-B–IV-D. Furthermore, it will be of great interest to explore more effective and efficient strategies to accelerate the optimization of the multiobjective mapping model. As MOEAs are population-based optimization methods, they have an implicit parallel processing capability, making parallel computing a potential way to improve the mapping efficiency.

V. CONCLUSION

In this article, we have proposed the novel MOMSPM method, in which the constrained optimization problem of spatiotemporal SPM is converted into an MOP with three objective functions. Globally, the MOEA/D method is used to simultaneously optimize the abundance, spatial, and temporal constraints and fuse their information to generate a well-balanced class distribution map. Locally, the GSDP and MAP operators are employed to refine the solution and achieve an improved outcome. The GSDP local search operator can improve the spatial correlation between subpixels, and the MAP-based local search operator can accelerate the convergence of the global evolution process. However, the GSDP local search operator can be less effective in handling larger abundance errors, due to the principle of the PSSM algorithm, and the MAP-based local search operator is less affected by abundance errors. Compared with the latest spatiotemporal SPM method of SST_MAP, MOMSPM can get rid of the sensitive weights. One synthetic dataset and two real-data datasets were used in the experiments. The results proved that the temporal information is helpful for the construction of the spatial distribution details. The experimental results also confirmed the superiority of the proposed methods in eliminating the errors of the abundance maps and producing an optimal solution.

However, the limitation of the proposed methods is their computational cost, as evolutionary algorithms do tend to require a lot of computation, especially for high-dimensional image data. Parallel computing to accelerate the algorithm will be a viable strategy that we will explore in the future.

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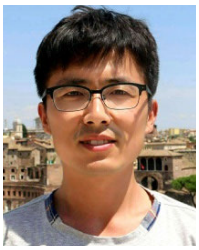
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